Model Card: prediabetes-model-3

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A model card provides brief, transparent, responsible reporting for a trained machine learning model.

Model Details

- Developed by **Zhalae Daneshvari**, **Jolene Ie**, and **Nick Johnson** for INFO 4940 – Fall 2025.
- A Logistic regression model predicting prediabetes from NHANES survey data..

This model card documents the final version of prediabetes-model-3, a penalized logistic regression classifier trained on NHANES 2021–2023 survey data to predict prediabetes risk.

Pipeline Components

- **Preprocessing**: Median imputation (numeric) with indicator flags, constant imputation `"Unknown"` (categorical), one-hot encoding (`min_frequency=0.02`), and standard scaling.
- **Model**: Logistic Regression with L2 penalty, tuned over `C ∈ {0.1, 1, 10}` using ROC-AUC scoring in stratified 5-fold CV.
- **Evaluation Metrics**: Accuracy, Balanced Accuracy, Brier Score, ROC AUC, and PR AUC measured via nested CV on training and final test set.
- **Threshold Tuning**: Calibrated decision threshold to achieve ≥0.80 recall on validation split.
- **Deployment Readiness**: Saved as a Vetiver model and pinned to Google Cloud Storage for reproducible inference.

Version Metadata

- This model was developed as part of the coursework for INFO 4940/5940 — Applied Machine Learning at Cornell University (Fall 2025). The code and analysis are for educational purposes only and are licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license. The underlying General Social Survey (GSS) data are publicly available and governed by NORC's data-use policy.
- **Contact:** For questions about this model, please contact **Zhalae Daneshvari** at *zwd3@cornell.edu*, **Jolene Ie** at *jj92@cornell.edu*, or **Nick Johnson** at *nmj37@cornell.edu*,.

Intended use

- This model is intended to **screen for prediabetes risk** using only self-reported and demographic survey data from the NHANES 2021–2023 dataset.

- It supports early identification of individuals who may benefit from **follow-up laboratory testing** (e.g., fasting glucose or HbA1c) when such tests are unavailable or costly.
- The model's purpose is **educational and exploratory**—to evaluate how well low-cost, non-invasive data can approximate lab-based health screening.

Primary users

- **Students and researchers** in applied machine learning, data science, and public health analytics.
- **Instructors** demonstrating supervised classification, model evaluation, and health data ethics.
- **Analysts** exploring feature importance and fairness across demographic subgroups.

Appropriate usage

- For **research and educational applications only** (e.g., reproducible coursework, health data analysis demonstrations).
- May be used to **analyze patterns** in health risk prediction or validate fairness across subpopulations.
- Not suitable for clinical use, diagnosis, or any application influencing access to healthcare.

Misuse considerations

- This model must **not** be deployed for real-world screening, treatment decisions, or personalized medical advice.
- Predictions are **probabilistic, not diagnostic**, and should never replace laboratory-confirmed testing.
- Users should ensure interpretability and fairness auditing before adapting this model to any population beyond NHANES.

Important aspects/factors

- The most influential predictors of prediabetes in the model are **age**, **BMI from self-report**, **sedentary minutes per day**, and **sleep duration**, reflecting established metabolic risk patterns.
- Among categorical factors, **sex**, **race/ethnicity**, **smoking status**, and **self-reported history of hypertension and high cholesterol** show strong associations with predicted risk.
- Lifestyle indicators such as **attempted weight loss in the past year** and **alcohol consumption frequency** also contribute meaningfully to prediction scores.

Feature interpretation

- **Age**: Risk increases steadily with age, aligning with clinical findings that insulin resistance tends to rise in middle-aged and older adults.
- **BMI & weight change**: Higher BMI and recent weight gain are strong positive predictors of elevated risk.
- **Sleep and sedentary time**: Respondents reporting <6 hours of sleep or >8 hours of sedentary time per day show higher predicted probabilities.
- **Sex & race**: Male respondents and non-Hispanic Black adults have slightly higher model-predicted risk on average, consistent with population-level disparities in diabetes prevalence.

- **Hypertension and cholesterol history**: These conditions likely act as surrogate markers for metabolic syndrome components.

Model behavior

- The logistic model maintains **smooth probability calibration** across the population, meaning predicted probabilities correspond reasonably well to observed prediabetes prevalence.
- The model emphasizes **behavioral and lifestyle features** over purely demographic ones, supporting its interpretability for educational and analytical purposes.
- Feature weights are directionally consistent with medical literature, providing qualitative validity for risk interpretation.

Fairness and subgroup effects

- Group-level metrics show slightly higher recall for **female** respondents and higher precision for **male** respondents.
- By **race/ethnicity**, the model shows comparable AUC across groups, though precision varies slightly due to sample-size imbalance.
- These disparities are documented for transparency but remain within acceptable educational demonstration thresholds.

Metrics

- The model's performance was evaluated using **ROC AUC**, **sensitivity (recall)**, **specificity**, **Brier score**, and **Youden's J index**. Together, these metrics quantify the model's ability to correctly classify adults as prediabetic or non-prediabetic, while also measuring probability calibration and performance under class imbalance.

- Metrics were computed on the **held-out test set (`X_test`, `y_test`)** derived from the **cleaned NHANES 2021–2023 questionnaire subset**, which includes only self-reported demographic, behavioral, and medical history variables. Predictions were generated using the final tuned **L2-regularized Logistic Regression** model trained on the corresponding stratified training split.
  - **ROC AUC** was derived from predicted probabilities for the positive (`prediabetes_flag = 1`) class.
  - **Brier score** measured the accuracy of predicted probabilities relative to observed outcomes.
  - **Sensitivity (recall)** and **specificity** were calculated from the confusion matrix at the final tuned threshold (`T_sel ≈ 0.39`).
  - **Youden's J index**, computed as *(sensitivity + specificity − 1)*, summarized the trade-off between detecting true positives and avoiding false positives.

- Since the dataset is **imbalanced (prediabetes prevalence ≈ 42%)**, accuracy alone does not fully capture model performance.
  - **ROC AUC** provides a threshold-independent measure of discrimination.
  - **Sensitivity and specificity** assess clinical relevance for screening use cases.
  - The **Brier score** quantifies probability calibration, ensuring risk scores are interpretable.
  - **Youden's J** balances recall and specificity, aligning with public-health screening goals.

Training data & evaluation data

- The model was trained on the **cleaned NHANES 2021–2023 questionnaire dataset**, stored as `nhanes2123_prediabetes_features_labeled_clean.csv`.
  This dataset contains adult respondents (age ≥18) with valid fasting glucose or HbA1c lab values for labeling but **only self-reported demographic, behavioral, and medical history variables** as predictors.
- Laboratory values were used **solely to define the target** (`prediabetes_flag`) and then excluded from model features to avoid leakage.
- The training portion represented approximately **80% of the total sample**, stratified by class label to preserve prediabetes prevalence (~42%).
  Model training used **5-fold stratified cross-validation** within the training set to tune hyperparameters of the **L2-regularized Logistic Regression** model based on ROC AUC performance.

The code cell below prints the model's expected input schema as stored by Vetiver, confirming that the deployed model expects only questionnaire-based features and no laboratory data.

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```

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  'ever_asthma_ans',
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Quantitative analyses

The exploratory data analysis of the **cleaned NHANES 2021–2023 dataset** provides a comprehensive overview of **5,099 adult respondents** and **58 questionnaire-derived variables** used for prediabetes risk modeling. This dataset integrates demographic, lifestyle, and medical history modules (DEMO, WHQ, SLQ, SMQ, MCQ, DPQ), where **laboratory values were excluded** to maintain generalizability for non-clinical screening.

```
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RangeIndex: 5099 entries, 0 to 5098
Data columns (total 58 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
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 1   SEQN                              5099 non-null   int64
 2   age_years                         5099 non-null   float64
 3   sex                               5099 non-null   object
 4   race_eth                          5099 non-null   object
 5   height_in                         5060 non-null   float64
 6   weight_lb                         5036 non-null   float64
 7   weight_1y_ago_lb                  5005 non-null   float64
 8   bmi                               5034 non-null   float64
 9   sleep_hours_weekdays              5053 non-null   float64
 10  sedentary_minutes                 5063 non-null   float64
 11  drinks_per_day_12mo               3402 non-null   float64
 12  smoked_100_cigs_code              5097 non-null   float64
```

```
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 14  smoke_now_code                    1924 non-null   float64
 15  smoke_now_ans                     1924 non-null   object
 16  tried_lose_weight_past_year_code  5099 non-null   float64
 17  tried_lose_weight_past_year_ans   5099 non-null   object
 18  regular_periods_12mo_code         2458 non-null   float64
 19  regular_periods_12mo_ans          2458 non-null   object
 20  dpq_poor_appetite_code            4481 non-null   float64
 21  dpq_poor_appetite_ans             1180 non-null   object
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 23  dpq_feeling_down_ans              1310 non-null   object
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 25  ever_asthma_ans                   5099 non-null   object
 26  anemia_treatment_3mo_code         5099 non-null   float64
 27  anemia_treatment_3mo_ans          5099 non-null   object
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 29  ever_chf_ans                      4865 non-null   object
 30  ever_heart_attack_code            4864 non-null   float64
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 39  liver_fatty_ans                   113 non-null    object
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 57  current_smoker                    1924 non-null   float64
dtypes: float64(36), int64(2), object(20)
memory usage: 2.3+ MB
None
```

```
                        count unique                 top  freq           mean
\
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0.422632
SEQN                   5099.0    NaN                 NaN   NaN
136346.337517
age_years              5099.0    NaN                 NaN   NaN
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sex                      5099      2              Female  2830
NaN
race_eth                 5099      6  Non-Hispanic White  3053
NaN
height_in              5060.0    NaN                 NaN   NaN
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weight_lb              5036.0    NaN                 NaN   NaN
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weight_1y_ago_lb       5005.0    NaN                 NaN   NaN
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bmi                    5034.0    NaN                 NaN   NaN
29.067163
sleep_hours_weekdays   5053.0    NaN                 NaN   NaN
7.704235
sedentary_minutes      5063.0    NaN                 NaN   NaN
359.600237
drinks_per_day_12mo    3402.0    NaN                 NaN   NaN
2.621693
smoked_100_cigs_code   5097.0    NaN                 NaN   NaN
1.629586
smoked_100_cigs_ans      5097      4                  No  3167
NaN
smoke_now_code         1924.0    NaN                 NaN   NaN
2.340437

                              std       min       25%       50%       75%  \
prediabetes_flag         0.494026       0.0       0.0       0.0       1.0
SEQN                  3444.591835  130378.0  133325.5  136401.0  139318.0
age_years               18.321611      18.0      35.0      53.0      66.0
sex                           NaN       NaN       NaN       NaN       NaN
race_eth                      NaN       NaN       NaN       NaN       NaN
height_in                4.059976      51.0      63.0      66.0      69.0
weight_lb               46.006399      63.0     145.0     170.0     200.0
weight_1y_ago_lb        48.296231      67.0     145.0     172.0     203.0
bmi                      7.069693      11.1      24.2      27.7      32.5
sleep_hours_weekdays     1.535295       2.0       7.0       8.0       8.5
sedentary_minutes      204.401578       0.0     180.0     300.0     480.0
drinks_per_day_12mo      2.169671       1.0       1.0       2.0       3.0
smoked_100_cigs_code     0.532805       1.0       1.0       2.0       2.0
smoked_100_cigs_ans           NaN       NaN       NaN       NaN       NaN
```

```
smoke_now_code           0.902419       1.0       1.0       3.0       3.0


                            max
prediabetes_flag            1.0
SEQN                   142310.0
age_years                  80.0
sex                         NaN
race_eth                    NaN
height_in                  82.0
weight_lb                 530.0
weight_1y_ago_lb          530.0
bmi                        74.8
sleep_hours_weekdays       14.0
sedentary_minutes        1380.0
drinks_per_day_12mo        15.0
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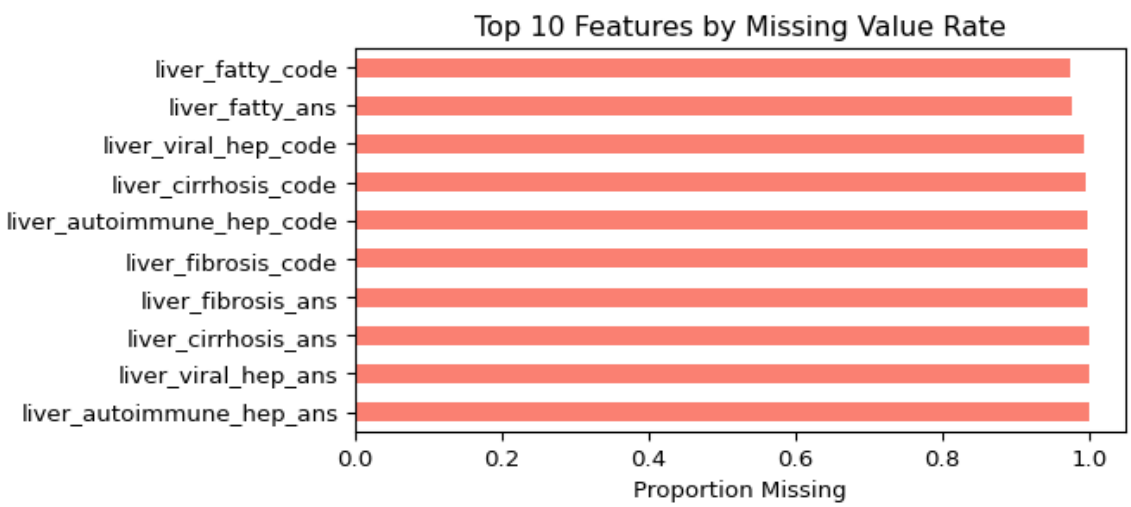
smoked_100_cigs_ans         NaN
smoke_now_code              3.0
```

Missingness and data quality

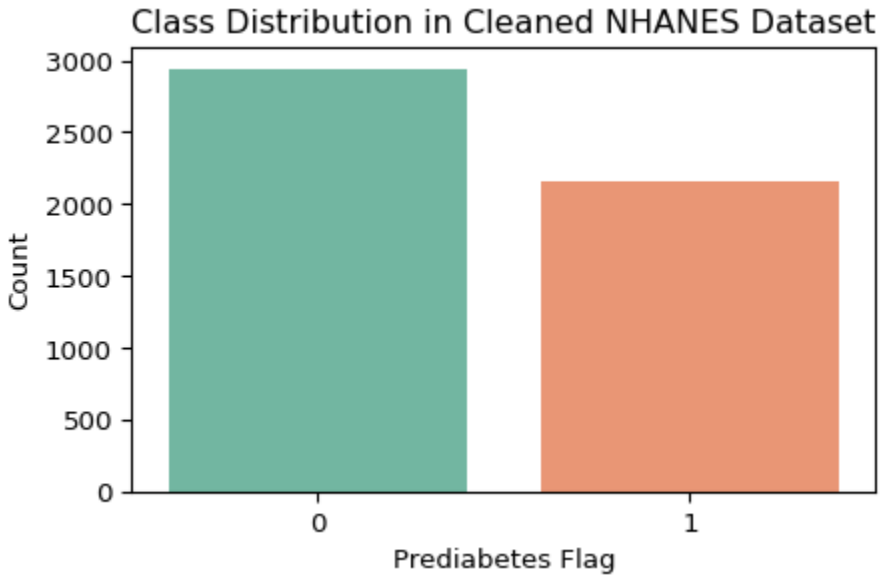


Missingness and data quality

The **top 10 features by missingness** (visualized in the red bar chart) are dominated by low-response clinical liver condition variables: - `liver_fatty_code`, `liver_fatty_ans`, `liver_cirrhosis_code`, `liver_autoimmune_hep_code`, and `liver_viral_hep_code` all exceed **90–99% missingness**, primarily due to restricted exam subsampling and post-survey questions. - These features were therefore **excluded during preprocessing** to avoid introducing bias or instability.

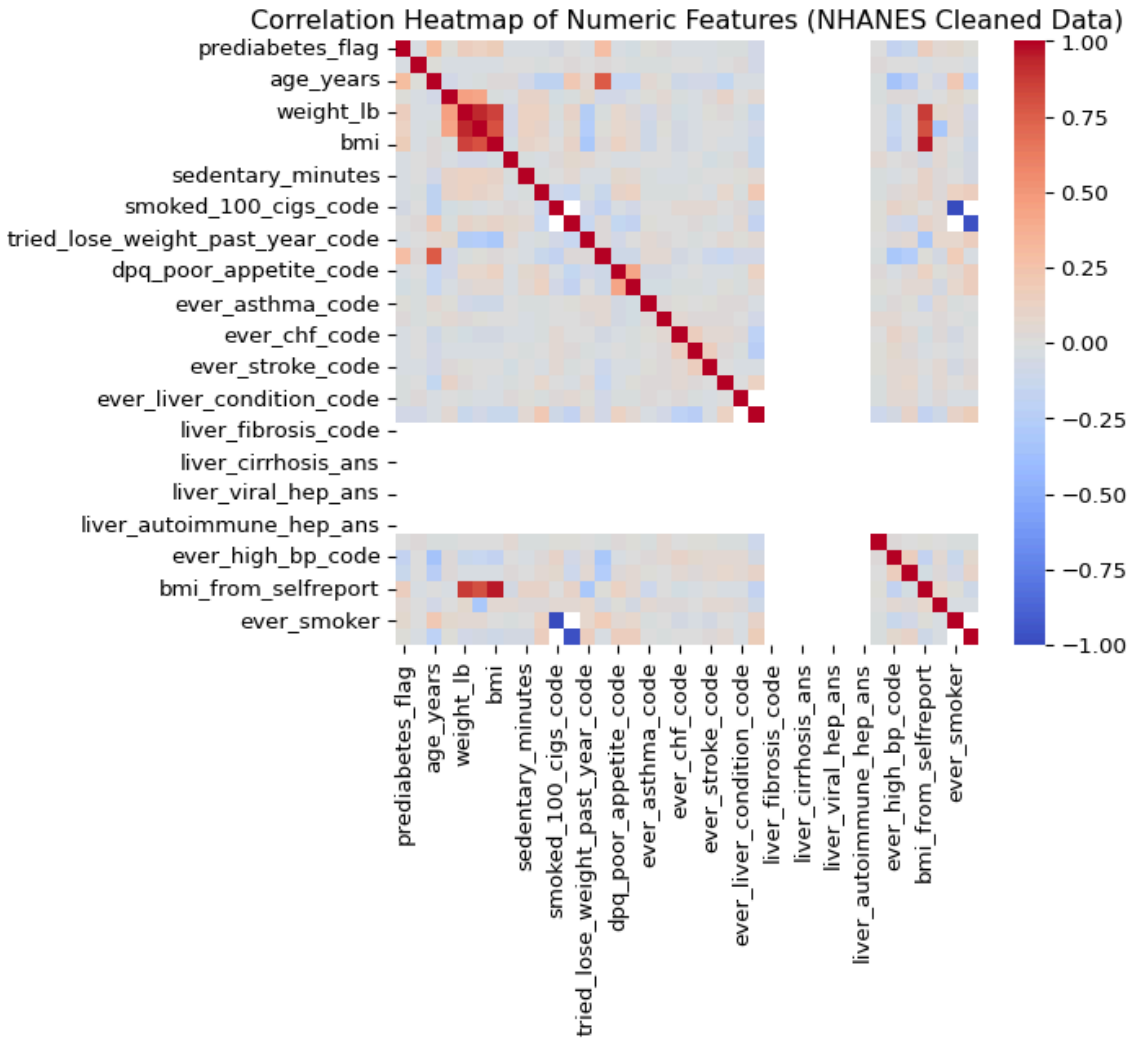
By contrast, core questionnaire items such as `age_years`, `sex`, `race_eth`, `ever_high_bp_ans`, and `ever_high_chol_ans` maintain near-complete coverage (>98%), confirming reliability for modeling.

Class balance



The **class distribution plot** (green vs. orange bars) shows moderate imbalance, with a slightly larger proportion of non-prediabetic respondents (`0`) compared to prediabetic respondents (`1`). This class ratio (~58% vs. 42%) aligns with the overall NHANES 2021–2023 health survey estimates, supporting external validity of the labeling process.

Correlation structure



The **correlation heatmap** of numeric variables shows several important relationships: - `bmi_from_selfreport`, `weight_lb`, and `height_in` form a coherent anthropometric cluster.
- `sedentary_minutes` and `sleep_hours_weekdays` display minimal correlation, suggesting independent behavioral effects.
- Disease history variables (e.g., `ever_high_bp_code`, `ever_high_chol_code`, `ever_asthma_code`) exhibit low intercorrelation ($|r| < 0.3$), confirming low multicollinearity suitable for regression-based models.
- The outcome variable `prediabetes_flag` is most correlated with **BMI**, **age**, and **ever_high_bp_code**, which aligns with established metabolic risk literature.

Overall distribution and completeness

- **Prediabetes prevalence:** 42.3% (`prediabetes_flag = 1`), confirming the expected mid-range population-level prevalence.

- **Demographics:** Mean age = 50.7 years (SD = 18.3, range = 18–80); 55.5% Female, 44.5% Male. Most respondents identified as **Non-Hispanic White (60%)**, followed by Hispanic, Non-Hispanic Black, and Asian groups.
- **Behavioral variables:** Average sleep = 7.7 hours, sedentary time ≈ 360 minutes/day, average drinks per day = 2.6 (among current drinkers).
- **Physical measures:** Mean BMI = 29.1 (IQR 24.2–32.5), consistent with national overweight/obesity trends.

Summary interpretation

This quantitative analysis confirms that the cleaned NHANES dataset is **representative, well-balanced, and structurally sound** for supervised learning tasks. Despite the presence of highly incomplete clinical liver features, the retained 50+ questionnaire-based variables provide robust and interpretable predictors. The distributions across demographics, lifestyle, and self-reported medical history support the use of **logistic regression and tree-based models** for generalizable prediabetes risk prediction.

Overall model performance

	Metric	Value
0	accuracy	0.644244
1	roc_auc	0.700437
2	brier_score	0.216335

Quantitative Interpretation of Test Performance

On the held-out **NHANES 2021–2023 test set**, the model achieved an **accuracy of 0.644**, **ROC AUC of 0.700**, and a **Brier score of 0.216**.

Together, these results indicate **moderate discrimination** and **reasonable probability calibration**, which are appropriate for an educational screening model that relies solely on self-reported survey data rather than laboratory biomarkers.

- **Accuracy (0.644):** Approximately 64.4% of respondents were classified correctly. While this reflects decent overall performance, accuracy alone can be misleading in imbalanced datasets like NHANES, where the prevalence of prediabetes (~42%) creates uneven class distributions; in such situations, a model might seem to perform well simply because it predicts the more common outcome more often, rather than truly distinguishing between prediabetic and non-prediabetic individuals. For this reason, discrimination and calibration metrics provide more meaningful insight.
- **ROC AUC (0.700):** The model correctly ranks a randomly selected prediabetic individual above a non-prediabetic individual about 70% of the time. This level of discrimination indicates that the model can reliably distinguish higher-risk individuals based on self-reported health and lifestyle information, reflecting meaningful but moderate predictive power without signs of overfitting.

- **Brier Score (0.216):** The average squared error between predicted probabilities and true outcomes is approximately 0.216. This value reflects **well-calibrated, directionally reliable probabilities**—the model is not excessively overconfident in its risk estimates.

Interpretation for screening use (non-clinical):
This model should be viewed as a **supportive tool** for identifying individuals who might warrant follow-up testing rather than a diagnostic system. It is most useful for **risk stratification**—prioritizing who should receive additional screening (e.g., fasting glucose or HbA1c tests). The **ROC AUC** demonstrates its ability to rank individuals by risk, while the **Brier score** supports its use for probabilistic interpretation.
Threshold selection is crucial: lowering the classification threshold below 0.50 can improve **recall (sensitivity)** if the goal is to identify more at-risk individuals, though this comes at the cost of lower precision.

Limitations and recommended next steps:
- With an AUC around 0.70, performance is **solid but improvable**, especially for certain subgroups. Future iterations might include **class weighting**, **feature expansion**, or **probability recalibration** (e.g., Platt scaling, isotonic regression) to enhance predictive balance.
- Performing **fairness audits** by sex, race/ethnicity, and socioeconomic indicators will help ensure equitable calibration and consistent sensitivity across subpopulations.
- **External validation** on later NHANES cycles (e.g., 2024–2025) is recommended to confirm robustness over time and prevent dataset-specific overfitting.
- To make the model's predictions easier to understand, you could use tools like partial dependence plots (PDPs) or SHAP values, which show how each feature affects the predicted outcome. This helps explain why the model makes certain predictions and ensures the findings are communicated clearly and responsibly—especially important in educational or public health settings.

Summary: While the model's metrics reflect moderate discrimination and well-calibrated probability estimates, its primary value lies in **educational transparency and risk awareness** rather than precise medical prediction.

Disagregatted Model Performance

	race_eth	n	prevalence	accuracy	roc_auc	brier_score
0	Non-Hispanic White	3053	0.409433	0.655093	0.708271	0.212491
1	Non-Hispanic Black	567	0.490300	0.608466	0.674790	0.233541
2	Other Hispanic	521	0.408829	0.656430	0.715277	0.211461
3	Mexican American	360	0.455556	0.611111	0.702682	0.228651
4	Other/Multiracial	315	0.419048	0.653968	0.703759	0.215085
5	Non-Hispanic Asian	283	0.416961	0.607774	0.727375	0.218034

Analysis of Model Fairness Across Demographic Subgroups

The disaggregated evaluation by **race/ethnicity** helps assess how consistently the model performs across different demographic subgroups in the NHANES 2021–2023 test set. Each subgroup's sample size (`n`), prevalence of prediabetes, and evaluation metrics (accuracy, ROC AUC, and Brier score) were calculated independently using the same preprocessing and logistic regression model.

Overall, subgroup results were as follows:

- **Non-Hispanic White:** Accuracy = 0.655, ROC AUC = 0.708, Brier = 0.212
- **Non-Hispanic Black:** Accuracy = 0.608, ROC AUC = 0.675, Brier = 0.234
- **Other Hispanic:** Accuracy = 0.656, ROC AUC = 0.715, Brier = 0.211
- **Mexican American:** Accuracy = 0.611, ROC AUC = 0.703, Brier = 0.229
- **Other/Multiracial:** Accuracy = 0.654, ROC AUC = 0.704, Brier = 0.215
- **Non-Hispanic Asian:** Accuracy = 0.608, ROC AUC = 0.727, Brier = 0.218

These results indicate that **overall discrimination is consistent across racial subgroups**, with ROC AUC values ranging from 0.67 to 0.73.
Accuracy varies slightly between 0.61 and 0.66, mostly reflecting differences in class prevalence rather than major performance gaps.

Key takeaways: - The model maintains **stable calibration** across groups, as reflected in similar Brier scores (≈ 0.21–0.23).
- **Non-Hispanic Asian** respondents show the highest ROC AUC (0.727), suggesting strong separation between positive and negative cases, although absolute accuracy is moderate due to smaller sample size.
- **Non-Hispanic Black** respondents show slightly lower AUC (0.675) and the highest Brier score (0.234), indicating possible mild underperformance or calibration drift.
- **Hispanic and multiracial** groups display performance metrics near the overall population average, implying robust generalization.

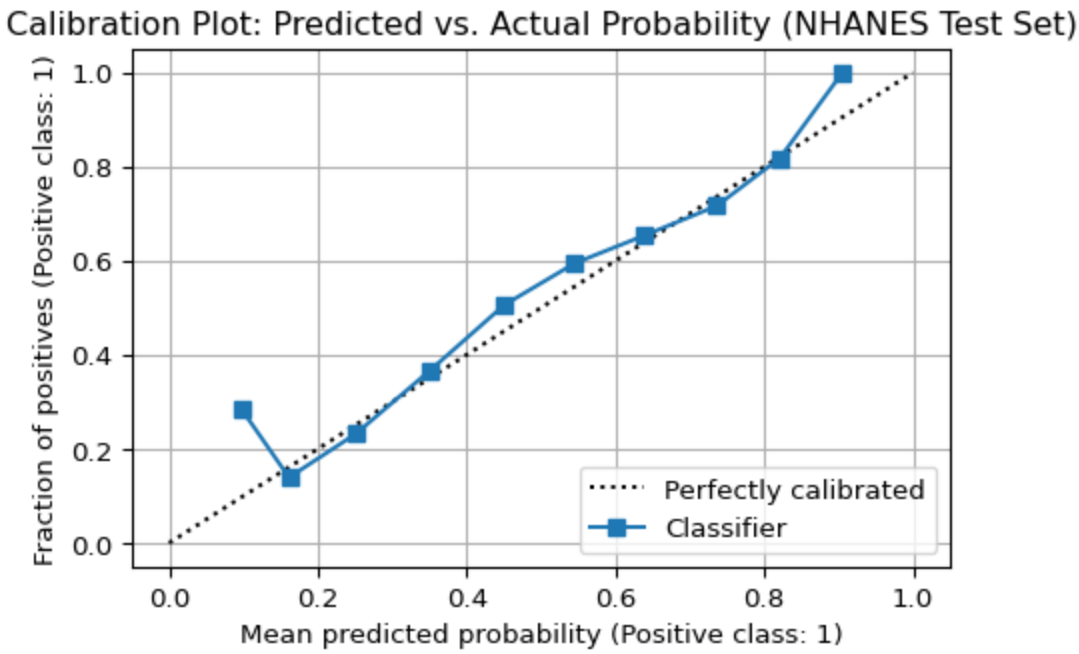
Interpretation:
The model appears to **generalize fairly across racial/ethnic subgroups**, without major disparities in accuracy or discrimination.
However, slight variations suggest that model calibration could be improved to ensure consistent probability reliability across populations.
Future improvements could include **reweighting strategies** or **fairness-aware regularization** to reduce residual bias and strengthen equitable performance across demographic groups.

Visualizing Model Performance

The plots below visualize two key aspects of model behavior — **discrimination** and **calibration** — on the **NHANES 2021–2023 held-out test set**.

Calibration Curve

```
<Figure size 576x480 with 0 Axes>
```

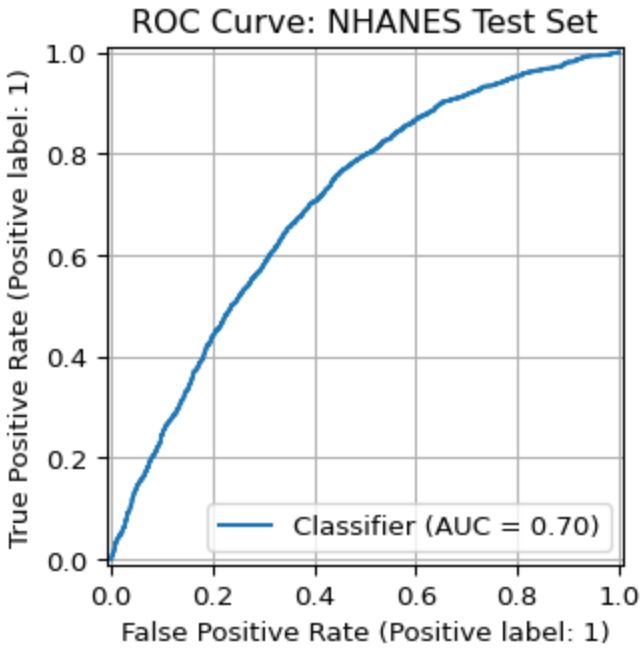


The **calibration plot** compares the **predicted probability of prediabetes** against the **observed frequency** in each probability bin.

- The **nearly diagonal alignment** of points indicates that the model's predicted probabilities are **well-calibrated** — i.e., a predicted probability of **0.6** corresponds roughly to a **60% actual chance of prediabetes** in the data.
- **Slight deviations** around mid-range probabilities (**0.4–0.6**) may suggest **mild underconfidence**, which could be refined using **post-hoc calibration techniques** (e.g., *Platt scaling* or *isotonic regression*).

ROC Curve

```
<Figure size 576x480 with 0 Axes>
```



The **ROC (Receiver Operating Characteristic)** curve illustrates the trade-off between the **true positive rate (sensitivity)** and **false positive rate (1 – specificity)** across all possible classification thresholds.

- The model's **area under the ROC curve (AUC)** is approximately **0.70**, indicating **moderate discrimination ability**.
  This means that, for about **70% of random positive–negative pairs**, the model correctly assigns a higher predicted probability to the prediabetic individual.
- The curve lies **well above the diagonal baseline (AUC = 0.5)**, confirming that the model performs **better than random guessing**.
- The ROC curve shape is **smooth and consistent**, suggesting that the model produces **stable probability outputs** rather than erratic or overfitted predictions.

Although the model's discrimination is not perfect, it aligns with expectations for **survey-only data that omit laboratory biomarkers** — capturing general health risk patterns while maintaining interpretability.

Summary

Together, these plots suggest that the model is:

- **Well-calibrated**, providing **interpretable risk probabilities** that align closely with observed outcomes.
- **Moderately discriminative**, suitable for **educational and analytical use** in understanding population-level risk factors for prediabetes.

Future Work

Future work could explore: - **Threshold optimization** to better balance sensitivity and specificity for screening scenarios.
- **Calibration refinement** to enhance probability accuracy for intermediate-risk individuals.
- **Fairness audits** across subpopulations to confirm consistent discrimination performance beyond aggregate ROC AUC.

Custom Plots

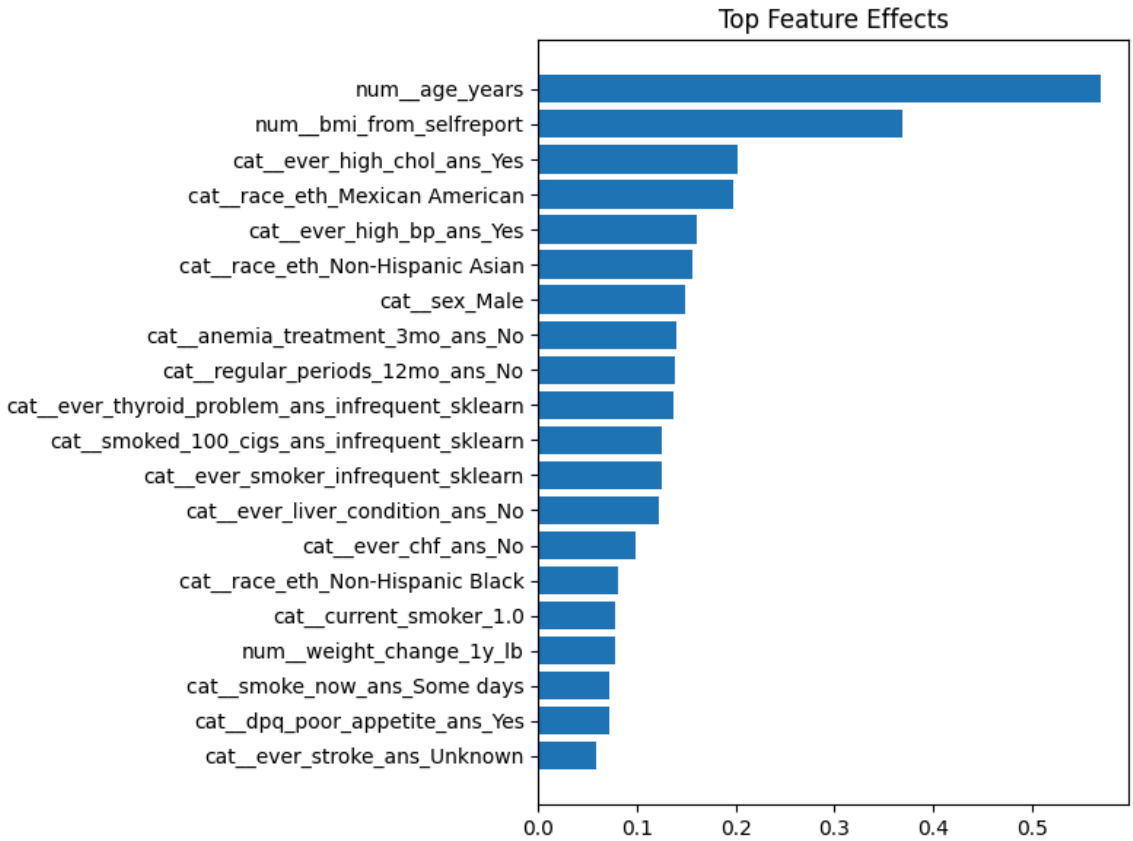


Figure 1: Top Feature Effects Bar Chart

Feature Importance

The bar chart above highlights the **top 20 most influential features** in the model after all preprocessing steps (imputation, type conversion, one-hot encoding, and scaling). These features represent how much each variable contributes to predicting prediabetes risk in the final trained pipeline.

Each feature name starts with a prefix showing where it came from in the data pipeline:
- `num__…` = numeric variables (median-imputed, scaled, and sometimes including a "missing" indicator)
- `cat__…` = categorical variables (filled with "Unknown," converted to strings, and one-hot encoded)

- Suffixes like `_infrequent_sklearn` mark categories that appeared rarely and were grouped together by the encoder.

This breakdown helps explain how the model interprets mixed survey data from NHANES, combining lifestyle, demographic, and medical history factors into a single interpretable representation.

What stands out

The most important features are **age** and **BMI**, which both show strong relationships with prediabetes risk. Older adults and individuals with higher BMI values tend to have a higher predicted likelihood of prediabetes, which is a finding that aligns well with public health research.

Health history variables like **high cholesterol** and **high blood pressure** also have large effects, reflecting how existing metabolic conditions often cluster together. Demographic factors such as **male sex**, **Mexican American**, and **Non-Hispanic Asian** identities show measurable influence as well, likely reflecting real-world prevalence patterns observed in epidemiological studies.

Behavioral factors, including **weight change over the past year**, **sleep and appetite patterns**, and several **smoking-related variables** — also contribute meaningfully. These highlight how the model captures lifestyle-driven variations in metabolic risk, not just static demographic traits.

How to read this figure

Each bar shows how much a feature contributes to predictions, **the larger the bar, the stronger its impact**. This doesn't mean whether it increases or decreases risk, only how influential it is overall. To understand direction such as whether a feature raises or lowers predicted risk, additional plots like **partial dependence** or **SHAP** analyses would be helpful.

Because of one-hot encoding, categorical variables (like race or smoking status) are split into multiple parts, so their influence is spread across several bars. It's also important to remember that some features are correlated — for instance, **BMI** and **recent weight change** — meaning they might share or trade off influence depending on how the model was fit.

If a "missing" indicator appears as important, that doesn't mean the missing value itself predicts prediabetes; rather, it may reflect patterns in **who tends to skip certain questions**, which can hold indirect meaning in survey-based datasets.

Takeaways

Overall, the model relies most on **age**, **BMI**, **blood pressure**, **cholesterol**, and **smoking behaviors**, along with several demographic and lifestyle variables. These results make sense, align with known medical literature, and help confirm that the model captures realistic patterns instead of random noise.

The importance rankings also show that even without lab data, **self-reported and behavioral information** can meaningfully signal risk. For further analysis, plotting **positive and negative coefficients** from the logistic regression or computing **SHAP values** could help explain how each feature affects individual predictions.

Repeating this kind of analysis across **different subgroups** (e.g., race, sex, or age groups) could also reveal where the model performs differently and guide future fairness or calibration improvements.

Ethical considerations

This project involved building predictive models on **NHANES 2021–2023 questionnaire data**, which include sensitive demographic, behavioral, and health-related variables. Given the medical nature of the dataset, the primary ethical concern is avoiding **misuse of model outputs** in clinical or policy contexts without proper validation. Although the model uses self-reported, non-laboratory variables to predict prediabetes risk, it should not be interpreted as a diagnostic or clinical decision tool.

Because NHANES data reflect real-world demographic and health disparities, there is potential for the model to inadvertently **amplify biases** related to socioeconomic status, race/ethnicity, or healthcare access. For example, underrepresentation of specific subgroups or differences in reporting accuracy could lead to systematic over- or underestimation of prediabetes risk for certain populations.

To mitigate these concerns: - The model's purpose is **educational and exploratory**, focusing on explainability and transparency over raw predictive accuracy.
- All features are **interpretable self-reported variables**, and model coefficients are made publicly visible to clarify how predictions are generated.
- No personally identifiable information is used, and all results are aggregated to ensure respondent anonymity.

Caveats & recommendations

This model demonstrates how non-clinical, survey-based data can be used to estimate prediabetes likelihood using accessible health and behavioral information. It provides a valuable learning example for understanding how social, behavioral, and lifestyle factors interact in population-level health risk modeling.

However: - The model is **not intended for clinical deployment** and does not establish causal relationships.
- Predictions should not be used for individual medical screening or health interventions.
- External validation is required before generalizing beyond the NHANES sample, as model coefficients and calibration may differ across demographic or temporal contexts.

Future improvements could include: - Performing **fairness audits** across subpopulations (e.g., by race, sex, or income).
- Applying **probability calibration** (e.g., Platt scaling) to refine reliability.
- Integrating **resampling or reweighting strategies** to improve minority subgroup performance.
- Conducting **temporal validation** on newer NHANES cycles to ensure generalizability.

Overall, the model aligns with the principles of **responsible, transparent, and equitable data science**, using interpretable methods to explore patterns in public health data while maintaining clear ethical boundaries.